

2025-2026 – Econ 0107 – Macroeconomics I

Lecture 4 : Fiscal Policies in a Growth Model

(Chapter 11 in LJUNQVIST & SARGENT)

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1. Introduction

- ▶ Complete market economy
- ▶ Time-0 trading
- ▶ Add production and taxes

2. The economy

2.1. Preferences, Technology, Information

- ▶ No uncertainty
- ▶ Representative household (hh)

$$\sum_{t=0}^{\infty} \beta^t U(c_t, 1 - n_t) \quad (2.1)$$

- ▶ Typically, in DSGEs:
 - × $U = u(c) + v(1 - n)$
 - × $U = \log c + \zeta \log(1 - n)$
 - × $U = \log c + \zeta \times (1 - n)$
 - × $U = u(c)$ (fixed labor supply)

2.1. Preferences, Technology, Information

► Technology:

$$F(k_t, n_t) \geq g_t + c_t + x_t \quad (2.2.a)$$

$$k_{t+1} = (1 - \delta)k_t + x_t \quad (2.2.b)$$

$\rightsquigarrow$

$$g_t + c_t + k_{t+1} \leq F(k_t, n_t) + (1 - \delta)k_t \quad (2.3)$$

- F is a neoclassical production function: linearly homogenous of degree 1:
 $F(\lambda k, \lambda n) = \lambda^1 F(k, n)$
- Euler theorem: $F_k k + F_n n = 1 \times F$
- Example: $F = k^\alpha n^{1-\alpha}$, $0 < \alpha < 1$

2.2. Components of a competitive equilibrium

- ▶ (Representative) Hh: owns capital, makes investment decisions, sells labour and capital services to the representative firm
- ▶ (Representative) Firm: rents labour and capital to produce final good
- ▶ price system $\{q_t, \eta_t, w_t\}$:
 - × pre-tax prices
 - × q_t (formerly denoted q_t^0): price of one unit of investment or consumption in t in units of time 0 numéraire.
 - × η_t : price of capital services in units of time t good
 - × w_t : price of labour services in units of time t good

2.2. Components of a competitive equilibrium

Definition 1

A govt expenditure and tax plan that satisfies the govt budget constraint is budget-feasible

- ▶ Competitive equilibria are indexed by alternative budget-feasible govt policies
- ▶ Govt budget constraint:

$$\sum_{t=0}^{\infty} q_t g_t \leq \sum_{t=0}^{\infty} q_t (\tau_{ct} c_t + \tau_{kt} (\eta_t - \delta) k_t + \tau_{nt} w_t n_t + \tau_{ht}) \quad (2.5)$$

- ▶ Note: if the govt was optimising, it would use only lump sum tax τ_h .

2.2. Components of a competitive equilibrium

- Hh budget constraint:

$$\sum_{t=0}^{\infty} q_t ((1 + \tau_{ct})c_t + (k_{t+1} - (1 - \delta)k_t)) \leq \sum_{t=0}^{\infty} q_t \left(\underbrace{\eta_t k_t - \tau_{kt}(\eta_t - \delta)k_t}_{(1 - \tau_{kt})\eta_t k_t + \tau_{kt}\delta k_t} + (1 - \tau_{nt})w_t n_t - \tau_{ht} \right) \quad (2.4)$$

- Note: depreciation allowance δk_t from gross rentals on capital.

3. Term structure of interest rates

- ▶ $\{q_t\}_{t=0}^{\infty}$ encodes the term structure of interest rates

$$q_t = q_0 \frac{q_1}{q_0} \frac{q_2}{q_1} \cdots \frac{q_t}{q_{t-1}} = q_0 m_{0,1} m_{1,2} \cdots m_{t-1,t}$$

- ▶ $m_{t,t+1} = \frac{q_{t+1}}{q_t}$ is the one-period discount factor between t and $t+1$

$$m_{t,t+1} = R_{t,t+1}^{-1} = \frac{1}{1 + r_{t,t+1}} \approx e^{-r_{t,t+1}}$$

- ▶ We can write

$$\begin{aligned} q_t &= q_0 e^{-r_{0,1}} e^{-r_{1,2}} \cdots e^{-r_{t-1,t}} \\ &= q_0 e^{-(r_{0,1} + r_{1,2} + \cdots + r_{t-1,t})} \\ &= q_0 e^{-tr_{0,t}} \end{aligned}$$

with

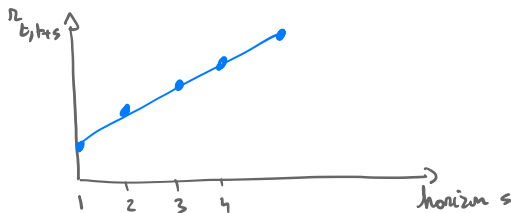
$$r_{0,t} = \frac{1}{t}(r_{0,1} + r_{1,2} + \cdots + r_{t-1,t})$$

3. Term structure of interest rates

- ▶ $q_t = q_0 e^{-tr_{0,t}}$
- ▶ $r_{0,t}$ is the net t -period rate of interest between 0 and t .
- ▶ It is the yield to maturity on a zero coupon bond that matures at t .
- ▶ More generally, one can write

$$r_{t,t+s} = \frac{1}{s}(r_{t,t+1} + r_{t+1,t+2} + \cdots + r_{t+s-1,t+s})$$

- ▶ From $s = 1, 2, \dots$, we obtain the yield curve



Detour: Interpreting the slope of the yield curve

- Take the simple endowment economy

$$\max \sum_t \beta^t \log c_t \quad \text{s.t.} \quad \sum_t q_t c_t \leq \sum_t q_t y_t \quad (\lambda)$$

- First order condition (*foc*) is $\beta^t \frac{1}{c_t} = \lambda q_t$
- Ratio of *foc* in $t + s$ and t :

$$\beta^s \frac{c_t}{c_{t+s}} = \frac{q_{t+s}}{q_t} = \frac{q_0 e^{-(t+s)r_{0,t+s}}}{q_0 e^{-(t)r_{0,t}}} = e^{-sr_{t,t+s}}$$

- Take the log and rearrange:

$$r_{t,t+s} = \gamma_{c_{t,t+s}} + \log \beta$$

where $\gamma_{c_{t,t+s}}$ is the average per period growth rate of consumption between t and $t + s$

- Expecting lower growth in the future implies that $r_{t,t+s}$ decreases with s (“inversion of the yield curve is a predictor of recession”)

Detour: Interpreting the slope of the yield curve

Opinion **On Wall Street**

What is the yield curve really telling us about the odds of recession?

Believers in predictive power of so-called inversion in bond markets still think it augurs downturn

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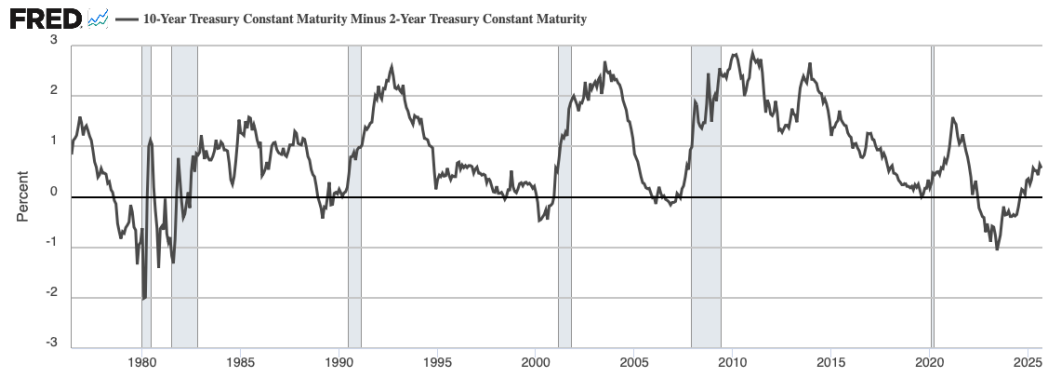


Before each of the six US recessions since 1980 hit, the curve inverted © George Widman/AP

Jennifer Hughes SEPTEMBER 21 2024

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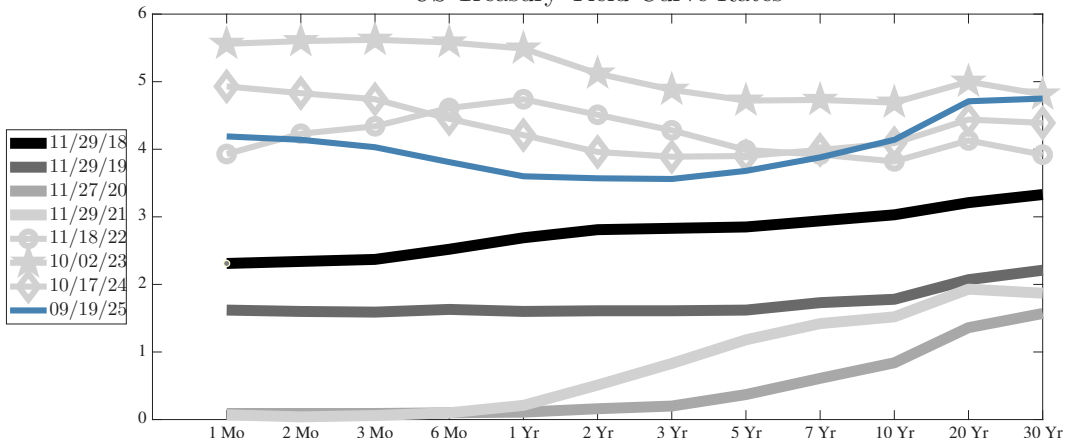
Detour: Interpreting the slope of the yield curve



Source: Federal Reserve Bank of St. Louis via FRED®
Shaded areas indicate U.S. recessions.

fred.stlouisfed.org

US Treasury Yield Curve Rates



4. Sequential version of the govt budget constraint

- ▶ It is useful to describe the sequence of on-period public debt associated with the expenditures and tax revenues (but it is not needed to compute the equilibrium)
- ▶ Assume no govt debt when entering period 0.
- ▶ Let T_t be the total tax revenues:

$$T_t = \tau_{ct}c_t + \tau_{kt}(\eta_t - \delta)k_t + \tau_{nt}w_t n_t + \tau_{ht}$$

The govt intertemporal budget constraint is

$$\sum_{t=0}^{\infty} q_t(g_t - T_t) = 0 \quad (4.1)$$

that can be rewritten as

$$\underbrace{g_0 - T_0}_{\text{current deficit}} = \underbrace{\sum_{t=1}^{\infty} \frac{q_t}{q_0} (T_t - g_t)}_{\text{discounted sum of future surpluses}} \quad (\star)$$

4. Sequential version of the govt budget constraint

- ▶ in a sequential world, one can think of period 0 deficit as being financed by debt B_0 :

$$B_0 = g_0 - T_0$$

- ▶ Therefore $(\star)$ implies

$$B_0 = \sum_{t=1}^{\infty} \frac{q_t}{q_0} (T_t - g_t)$$

or

$$\underbrace{\frac{q_0}{q_1}}_{R_{0,1}} B_0 = T_1 - g_1 + \underbrace{\sum_{t=2}^{\infty} \frac{q_t}{q_1} (T_t - g_t)}_{B_1}$$

or equivalently

$$g_1 + R_{0,1} B_0 = T_1 + B_1$$

4. Sequential version of the govt budget constraint

- In period t , we will have

$$g_t + R_{t-1,t}B_{t-1} = T_t + B_t$$

or

$$\underbrace{B_t - B_{t-1}}_{\text{new debt issuance}} = \underbrace{g_t - T_t}_{\text{primary deficit}} + \underbrace{r_{t-1,t}B_{t-1}}_{\text{net interest payments}} \quad (4.4)$$

- The Arrow-Debreu budget constraint (4.1) ensures the no-Ponzi scheme (transversality) condition

$$\lim_{t \rightarrow \infty} q_t B_{t+1} = 0$$

4. Sequential version of the govt budget constraint

► Note:

- × There is no loss of generality in considering only one-period debt
- × The maturity structure of govt debt is irrelevant.

5. Competitive equilibrium with distorting taxes

- ▶ Hh chooses $\{c_t, n_t, k_{t+1}\}$ for $t = 0, \dots$ to max U s.t. the budget constraint
- ▶ Firm chooses $\{k_t, n_t\}$ for $t = 0, \dots$ to max firm value V_0

$$V_0 = \sum_{t=0}^{\infty} q_t \underbrace{(F(k_t, n_t) - \eta_t k_t - w_t n_t)}_{\text{profit of period } t}$$

- ▶ A budget-feasible policy is an expenditure plan $\{g_t\}$ and a tax plan $\{\tau_{ct}, \tau_{nt}, \tau_{kt}, \tau_{ht}\}$ that satisfies the govt budget constraint.

5. Competitive equilibrium with distorting taxes

Definition 1

A competitive equilibrium with distorting taxes is

- ▶ *a budget-feasible allocation*
- ▶ *a feasible allocation*
- ▶ *a price system*

such that, given the price system and the govt policy

- ▶ *the allocation solves the hh problem*
- ▶ *the allocation solves the firm problem*

5.1. The hh: no-arbitrage condition and asset-pricing formula

- The hh intertemporal budget constraint (*ibc*) is

$$\sum_{t=0}^{\infty} q_t ((1 + \tau_{ct})c_t + (k_{t+1} - (1 - \delta)k_t)) \leq \sum_{t=0}^{\infty} q_t (\eta_t k_t - \tau_{kt}(\eta_t - \delta)k_t + (1 - \tau_{nt})w_t n_t - \tau_{ht}) \quad (2.4)$$

- Rewrite the terms in blue as follows (on the right-hand side of the *ibc*)

× terms in k_0 :

$$q_0(1 - \delta) + q_0\eta_0 - q_0\tau_{k0}(\eta_0 - \delta) = ((1 - \tau_{k0})(\eta_0 - \delta) + 1)q_0$$

× terms in k_t :

$$-q_{t-1} + q_t(1 - \delta) + q_t\eta_t - q_t\tau_{kt}(\eta_t - \delta) = \underbrace{((1 - \tau_{kt})(\eta_t - \delta) + 1)q_t}_{\text{return on capital}} - \underbrace{q_{t-1}}_{\text{cost of capital}}$$

5.1. The hh: no-arbitrage condition and asset-pricing formula

- Therefore, the budget constraint rewrites

$$\begin{aligned} \sum_{t=0}^{\infty} q_t(1 + \tau_{ct})c_t &\leq \sum_{t=0}^{\infty} q_t(1 - \tau_{nt})w_t n_t - \sum_{t=0}^{\infty} q_t \tau_{ht} \\ &\quad + \sum_{t=0}^{\infty} (((1 - \tau_{kt})(\eta_t - \delta) + 1)q_t - q_{t-1})k_t \\ &\quad + ((1 - \tau_{k0})(\eta_0 - \delta) + 1)q_0 k_0 \\ &\quad - \lim_{T \rightarrow \infty} q_T k_{T+1} \end{aligned} \quad (5.1)$$

- Hh would be happy to have the highest possible right-hand side of (5.1)
- But this rhs must be bounded in equilibrium (because resources are finite) $\rightsquigarrow$ this is putting restrictions on equilibrium prices.

5.1. The hh: no-arbitrage condition and asset-pricing formula

- ▶ Take the term in k_t : $\rho_t = \left(((1 - \tau_{kt})(\eta_t - \delta) + 1)q_t - q_{t-1} \right)$
 - × if $\rho_t > 0$: hh can:
 - ▶ buy in $t - 1$ arbitrarily large k_t with present value $q_{t-1}k_t$
 - ▶ sell in t rental services and undepreciated part to obtain a present value income of $((1 - \tau_{kt})(\eta_t - \delta) + 1)q_t k_t$
 - ▶ as $\rho_t > 0$, thus gives an arbitrarily large benefit
 - ▶ the rhs of (5.1) would then be unbounded $\leadsto$ not an equilibrium.
 - × if $\rho_t < 0$: hh can does the reverse:
 - ▶ short-sell in $t - 1$ at price q_{t-1}
 - ▶ deliver in t buy buying at price $((1 - \tau_{kt})(\eta_t - \delta) + 1)q_t$
 - ▶ again, the rhs of (5.1) would be unbounded $\leadsto$ not an equilibrium
- ▶ Therefore, by no-arbitrage

$$\frac{q_t}{q_{t+1}} = (1 - \tau_{kt+1})(\eta_{t+1} - \delta) + 1 \quad \forall t \geq 0 \quad (5.2)$$

- ▶ and no possibility to short-shell at $+\infty$:

$$\lim_{T \rightarrow \infty} q_T k_{T+1} = 0$$

5.2. User cost of capital

► Rewriting (5.2):

$$\underbrace{\eta_{t+1}}_{\text{user cost of capital}} = \underbrace{\delta}_{\text{depreciation}} + \underbrace{\left(\frac{1}{1 - \tau_{kt+1}} \right)}_{\text{taxes}} \underbrace{\left(\frac{q_t}{q_{t+1}} - 1 \right)}_{\text{capital gains or losses}} \quad (5.4)$$

5.3. Hh *foc*

$$\max \mathcal{L} = \sum_{t=0}^{\infty} \beta^t U(c_t, 1 - n_t) + \mu \text{ } ibc$$

- ▶ Hh are indifferent about the level of k_t as long as the no-arbitrage condition holds
- ▶ *foc* for c_t and n_t :

$$\beta^t U_{1t} = \mu q_t (1 + \tau_{ct}) \quad (5.5a)$$

$$\beta^t U_{2t} = \mu w_t (1 - \tau_{nt}) \quad (5.5b)$$

assuming an interior solution $n_t < 1$.

- ▶ We see that only μq_t matters, not μ and q_t separately $\rightsquigarrow$ once can choose a numéraire, or can arbitrarily normalize $\mu = 1$

5.4. A theory of the term structure of interest rates

- ▶ Assume $U(c_t, 1 - n_t) = u(c_t) + v(1 - n_t)$
- ▶ *foc* wrt to c_t :

$$\mu q_t = \beta^t \frac{u'(c_t)}{1 + \tau_{ct}}$$

- ▶ $\{q_t\}$ and therefore the term structure can be computed if we observe $\{c_t\} \rightsquigarrow$ CCAPM
- ▶ Govt policy $\{g_t, \tau_{ct}, \tau_{nt}, \tau_{kt}, \tau_{ht}\}$ affects equilibrium $\{c_t\}$, and therefore the term structure.

5.5. Firms

- Firm value is

$$V_0 = \sum_{t=0}^{\infty} q_t (F(k_t, n_t) - w_t n_t - \eta_t k_t)$$

- Because of homogeneity of degree 1 (Euler theorem):

$$V_0 = \sum_{t=0}^{\infty} q_t ((F_{nt} - w_t) n_t + (F_{kt} - \eta_t) k_t)$$

- By no-arbitrage:

$$\begin{aligned} \eta_t &= F_{kt} \\ w_t &= F_{nt} \end{aligned} \tag{5.7}$$

6. Computing equilibria

- ▶ $\{g_t, \tau_t\} = \{g_t, \tau_{ct}, \tau_{nt}, \tau_{kt}\}$ is exogenous
- ▶ $\sum_{t=0}^{\infty} q_t \tau_{ht}$ is endogenous and makes sure that the govt intertemporally balances its budget.

6.1. Inelastic labor supply

- ▶ assume $U(c, 1 - n) = u(c)$ and hh inelastically supply $n = 1$ (normalization)
- ▶ Define $f(k) = F(k, 1)$
- ▶ Feasibility writes

$$k_{t+1} = (1 - \delta)k_t + f(k_t) - g_t - c_t \quad (6.1)$$

- ▶ Note that $F(k, n) = nF(k/n, 1) = nf(\hat{k})$ with $k/n = \hat{k}$
- ▶ One then has:

$$F_k = \frac{\partial[nF(k/n, 1)]}{\partial k} = n \times \frac{1}{n} \times \frac{\partial F(k/n, 1)}{\partial(k/n)} = f'(\hat{k})$$

and

$$F_n = \frac{\partial[nF(k/n, 1)]}{\partial n} = n \times \frac{-k}{n^2} \times \frac{\partial F(k/n, 1)}{\partial(k/n)} + F(k/n, 1) = f(\hat{k}) - \hat{k}f'(\hat{k})$$

- ▶ and when $n = 1$, $F_k = f'(k)$ and $F_n = f(k) - kf'(k)$

6. Computing equilibria

Some substitutions

- Take resource constraint

$$k_{t+1} = (1 - \delta)k_t + f(k_t) - g_t - c_t$$

- Obtain c_t and replace in the *foc*

$$\beta^t u'(c_t) = \mu q_t (1 + \tau_{ct})$$

- Obtain q_t and q_{t+1} and replace in the no-arbitrage condition

$$\frac{q_t}{q_{t+1}} = (1 - \tau_{kt+1})(\eta_{t+1} - \delta) + 1$$

where η_{t+1} is replaced using the no-arbitrage condition $\eta_t = F_{kt}$

- We then obtain a nonlinear second order difference equation in k_t

6. Computing equilibria

A second order difference equation

$$\begin{aligned} \frac{u'(f(k_t + (1 - \delta)k_t - g_t - k_{t+1}))}{(1 - \tau_{ct})} &= \beta \frac{u'(f(k_{t+1} + (1 - \delta)k_{t+1} - g_{t+1} - k_{t+2}))}{(1 - \tau_{ct+1})} \\ &\times ((1 - \tau_{kt+1})(f'(k_{t+1}) - \delta) + 1) \\ &= 0 \end{aligned} \tag{6.2}$$

- ▶ initial condition k_0
- ▶ terminal condition $\lim_{T \rightarrow \infty} q_T k_{T+1} = 0$
- ▶ for given gvt policy
- ▶ (6.2) can be rewritten as

$$u'(c_t) = \beta u'(c_{t+1}) \frac{(1 - \tau_{ct})}{(1 - \tau_{ct+1})} ((1 - \tau_{kt+1})(f'(k_{t+1}) - \delta) + 1) \tag{6.3}$$

6.2. Equilibrium steady state

- ▶ Let $z_t = \{g_t, \tau_{kt}, \tau_{ct}\}$ be the sequence of exogenous variables
- ▶ (6.2) can be written as

$$H(k_t, k_{t+1}, k_{t+2}, z_t, z_{t+1}) = 0 \quad (6.4)$$

- ▶ For the steady state to be relevant, we look at cases where

$$\lim_{t \rightarrow \infty} z_t = \bar{z} \quad (6.5)$$

- ▶ At the steady state, we have

$$H(\bar{k}, \bar{k}, \bar{k}, \bar{z}, \bar{z}) = 0 \quad (6.6)$$

6.2. Equilibrium steady state

- (6.3) writes at the steady state

$$u'(\bar{c}) = \beta u'(\bar{c}) \frac{(1 - \bar{\tau}_c)}{(1 - \bar{\tau}_k)} ((1 - \bar{\tau}_k)(f'(\bar{k}) - \delta) + 1)$$

which gives

$$1 = \beta ((1 - \bar{\tau}_k)(f'(\bar{k}) - \delta) + 1) \quad (6.3b)$$

- Note: $\bar{\tau}_c$ does not distort $\bar{k}$
- With $\frac{1}{\beta} = 1 + \rho$, steady state capital is pinned down by

$$f'(\bar{k}) = \delta + \frac{\rho}{1 - \bar{\tau}_k}$$

6.3. Computing the equilibrium path with the shooting algorithm

- We want to solve the below difference equation system:

$$\begin{aligned}u'(c_t) &= \beta u'(c_{t+1}) \frac{(1 - \tau_{ct})}{(1 - \tau_{ct+1})} ((1 - \tau_{kt+1})(f'(k_{t+1}) - \delta) + 1) && \text{(Euler equation)} \\k_{t+1} &= (1 - \delta)k_t - g_t - c_t && (6.8a)\end{aligned}$$

with boundary conditions

$$\begin{cases} k_0 \text{ given} \\ \lim_{T \rightarrow \infty} \beta^T \frac{u'(c_T)}{(1 + \tau_{cT})} k_{T+1} \end{cases}$$

where we have used $\beta^T u'(c_T) = \mu q_t(1 + \tau_{ct})$

- Shooting algorithm:
 - × Take terminal period S large but finite
 - × We want to reach $k_S \approx \bar{k}$
 - × For given c_0 , iterate the difference system forward starting from (k_0, c_0) and compute k_S
 - × Try many values of c_0
 - × Solution is found for the c_0 such that $k_S \approx \bar{k}$

6.3. Computing the equilibrium path with the shooting algorithm

- Once this is done, find $\{\tau_{ht}\}$ such that the govt budget constraint is satisfied
- Then compute prices using

$$q_t = \beta^t \frac{u'(c_t)}{u'(c_{t+1})} \quad (6.8b)$$

$$\eta_t = f'(k_t) \quad (6.8c)$$

$$w_t = f(k_t) - k_t f'(k_t) \quad (6.8d)$$

$$\begin{aligned} \bar{R}_{t+1} &= \frac{1 + \tau_{ct}}{1 + \tau_{ct+1}} ((1 - \tau_{kt+1})(f'(k_{t+1}) - \delta) + 1) \quad (6.8e) \\ &= \frac{1 + \tau_{ct}}{1 + \tau_{ct+1}} R_{t,t+1} \end{aligned}$$

$$R_{t,t+1}^{-1} = m_{t,t+1} = \beta \frac{u'(c_t)}{u'(c_{t+1})} \frac{1 + \tau_{ct}}{1 + \tau_{ct+1}} \quad (6.8f)$$

$$r_{t,t+1} = R_{t,t+1} - 1 = (1 - \tau_{kt+1})(f'(k_{t+1}) - \delta) \quad (6.8g)$$

$$u'(c_t) = \beta u'(c_{t+1}) \bar{R}_{t+1} \quad (6.8h)$$

6.3. Computing the equilibrium path with the shooting algorithm

- ▶ if $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, then (6.8h) becomes

$$\log \frac{c_{t+1}}{c_t} = \gamma^{-1} \log \beta + \gamma^{-1} \log \bar{R}_{t+1} \quad (6.9)$$

- ▶ (6.9): consumption growth varies with distorted real interest rate.

6.6. When lump-sum taxes are available

- ▶ What we have just done is to implement the shooting algorithm taking as given $\{g_t, \tau_{ct}, \tau_{nt}, \tau_{kt}\}$.
- ▶ Then, once prices and quantities are obtained, $\{\tau_{ht}\}$ is set such that $\sum_{t=0}^{\infty} q_t \tau_{ht}$ balances the govt budget constraint.
- ▶ We can do this two-step computation because $\{\tau_{ht}\}$ are nowhere in equations (6.8)
- ▶ The timing of $\{\tau_{ht}\}$ is irrelevant $\rightsquigarrow$ Ricardian equivalence

6.7. When no lump-sum taxes are available

- ▶ Then, an additional step is needed in the algorithm: making sure that the govt budget constraint is satisfied.
- ▶ Algorithm given a sequence of $\{g_t\}$:
 - × Assume sequence of taxes $\{\tau_{ct}, \tau_{nt}, \tau_{kt}\}$
 - × solve for the equilibrium using the shooting algorithm
 - × check if the budget constraint of the govt is satisfied
 - × If not, adjust taxes and repeat.

8. Effect of taxes on equilibrium allocations and prices

- ▶ τ_c , τ_n and τ_k are distortionary, meaning that hh can affect their tax payments by altering their decisions.
- ▶ τ_h is non distortionary.

8.1. Lump-sum taxes and Ricardian equivalence

- ▶ Suppose $\tau_c = 0$, $\tau_n = 0$ and $\tau_k = 0 \rightsquigarrow \tau_h$ does not enter anywhere in (6.8)
- ▶ The timing of $\{\tau_{ht}\}$ is irrelevant, only $\sum q_t \tau_{ht}$ matters in govt and hh intertemporal budget constraints.
- ▶ This is Ricardian equivalence

8.2. When labour supply is inelastic

- ▶ τ_n is not distorting
- ▶ Constant τ_c is not distorting
- ▶ Variations in τ_c are distorting
- ▶ Capital taxation τ_k is distorting

9. Transition experiments with inelastic labour supply

- ▶ Assume

- × $U(c, 1 - n) = u(c) = \frac{c^{1-\gamma}}{1-\gamma}, f(k) = k^\alpha$

- × $\alpha = 1/3, \delta = 0.2, \beta = .95, g = 0.2$

- × $\gamma = 2$ or $\gamma = 0.2$

- ▶ First we do a foreseen once-for-all increase in g, τ_c, τ_k

- ▶ The change is announced at $t = 0$ and takes place at $t = 10$, and the economy was at the steady state before 0.

- ▶ Although no change is implemented before $t = 10$, the economy reacts on impact

- ▶ Why? Because hh wants to smooth consumption $\rightsquigarrow$ they adjust their savings from period 0 and onwards $\rightsquigarrow$ prices and quantities move at time 0.

- ▶ Two forces are at play in the dynamics:

- × discounting of the future before T

- × transient dynamics after T

and these two forces are interrelated (see later)

9. Transition experiments with inelastic labour supply

Foreseen permanent increase in g

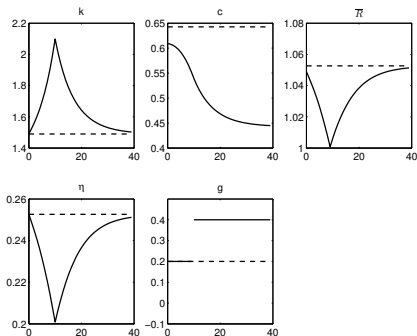
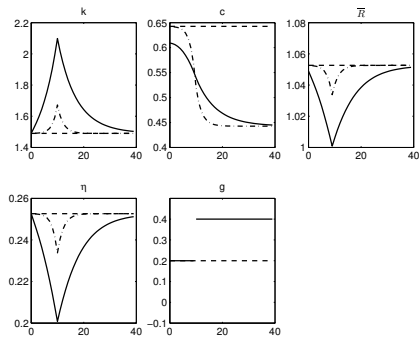


Figure 11.9.1: Response to foreseen once-and-for-all increase in g at $t = 10$. From left to right, top to bottom: $k, c, \bar{R}, \eta, g$. The dashed line is the original steady state.

- The steady state level of k is unaffected (see (6.3b))
- $g \nearrow \rightsquigarrow c \searrow$
- from 0 to 10: $c \searrow \rightsquigarrow k \nearrow$ (because $g \rightarrow$)
- Initial negative wealth effect of c (because $\sum q_t \tau_{ht} \nearrow$)
- The dynamics of $\bar{R}$ makes the hh choosing a non flat c profile.
- Both feedforward and feedback dimension in the response of the economy (more on this later)

9. Transition experiments with inelastic labour supply

Foreseen permanent increase in g , $\gamma = 2$ or 0.2



- More willingness to smooth consumption when $\gamma = 2$ as compared to when $\gamma = 0.2$
- When γ is small (the limit would be linear utility), c becomes the mirror image of g
- Less feedforward and less feedback effect $\rightsquigarrow$ the two dimensions are related (see later)

Figure 11.9.2: Response to foreseen once-and-for-all increase in g at $t = 10$. From left to right, top to bottom: $k, c, \bar{R}, \eta, g$. The dashed lines show the original steady state. The solid lines are for $\gamma = 2$, while the dashed-dotted lines are for $\gamma = .2$

9. Transition experiments with inelastic labour supply

Foreseen permanent increase in g , asset prices

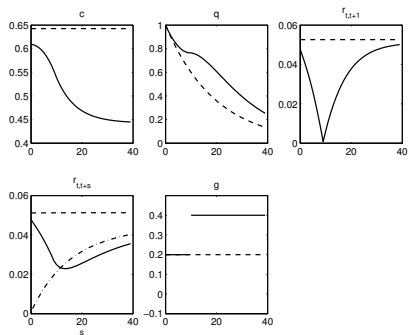


Figure 11.9.3: Response to foreseen once-and-for-all increase in g at $t = 10$. From left to right, top to bottom: $c, q, r_{t,t+1}$ and yield curves $r_{t,t+s}$ for $t = 0$ (solid line), $t = 10$ (dash-dotted line) and $t = 60$ (dashed line); term to maturity s is on the x axis for the yield curve, time t for the other panels.

- ▶ $q_t = \beta^t c_t^{-\gamma}$
- ▶ Short rate $r_{t,t+1} = -\log \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\gamma}$
- ▶ q_t : price of future consumption is higher in the future (when g is higher)
- ▶ Term structure at 10 periods: upward sloping because the growth rate of c is expected to increase (to be less negative)
- ▶ Term structure at time 0 is U shaped.

9. Transition experiments with inelastic labour supply

Foreseen permanent increase in τ_c

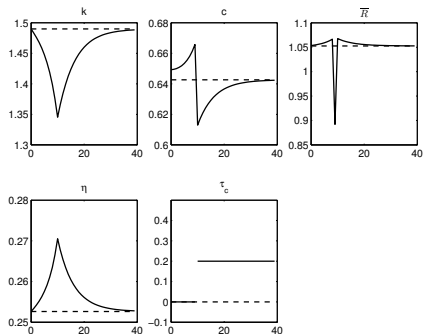
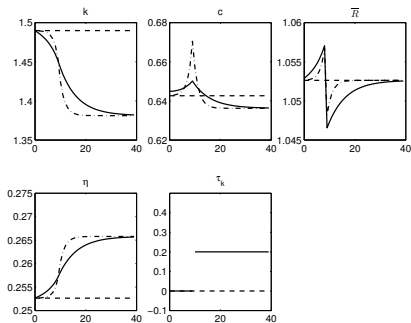


Figure 11.9.4: Response to foreseen once-and-for-all increase in τ_c at $t = 10$. From left to right, top to bottom: $k, c, \bar{R}, \eta, \tau_c$.

- ▶ $u'(c_t) = \beta u'(c_{t+1}) \frac{(1 - \tau_{ct})}{(1 - \tau_{ct+1})} \left((1 - \tau_{kt+1})(f'(k_{t+1}) - \delta) + 1 \right) \quad (6.3)$
- ▶ Anticipated decrease in $\frac{(1 - \tau_{ct})}{(1 - \tau_{ct+1})} \equiv$ anticipated increase in τ_k , as seen in (6.3)
- ▶ The hh frontloads consumption, by $\searrow k$
- ▶ No effect on the steady state
- ▶ After T , no more anticipation effect $\rightsquigarrow$ transient dynamics when starting with low k

9. Transition experiments with inelastic labour supply

Foreseen permanent increase in τ_k

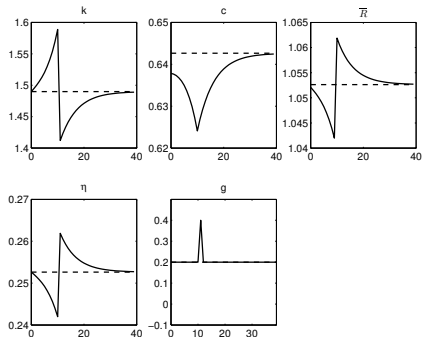


- Lower final steady state $\rightsquigarrow$ some capital can be eaten in the transition $\rightsquigarrow c \nearrow$ before period 10.
- After 10, transient dynamics from a higher than steady state stock of capital.

Figure 11.9.5: Response to foreseen increase in τ_k at $t = 10$. From left to right, top to bottom: $k, c, \bar{R}, \eta, \tau_k$. The solid lines depict equilibrium outcomes when $\gamma = 2$, the dashed-dotted lines when $\gamma = .2$.

9. Transition experiments with inelastic labour supply

One time impulse g_{10}



- Again, the anticipation effect is at play before 10
- Desire to smooth c
- in 10, govt takes out some good for g , but c stays smooth $\rightsquigarrow$ investment adjusts by $\searrow$.

Figure 11.9.6: Response to foreseen one-time pulse increase in g at $t = 10$. From left to right, top to bottom: $k, c, \bar{R}, \eta, g$.

10. Linear approximation

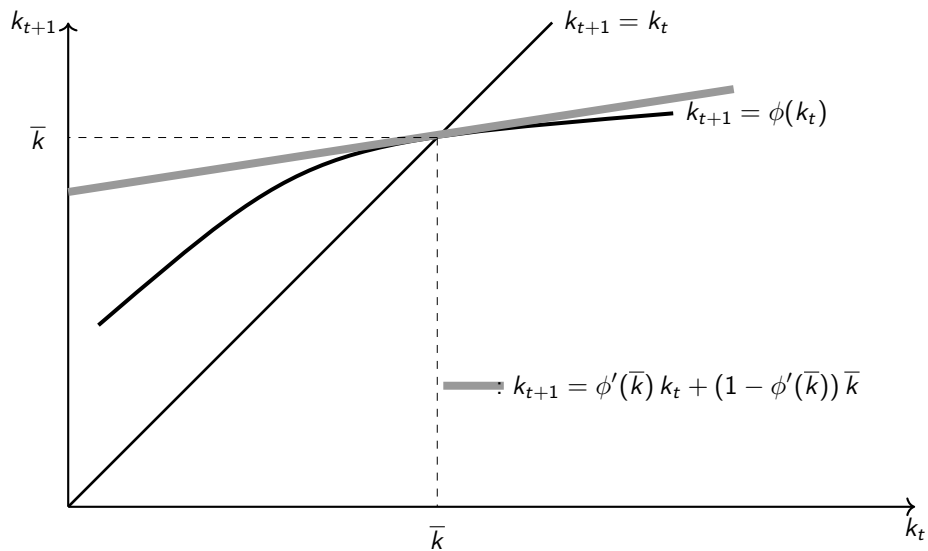
- ▶ Shooting algorithm can be tricky in larger models
- ▶ Useful to look at the solution of a linear approximation (one can also do log-linear)
- ▶ idea: Assume the model is

$$k_{t+1} = \varphi(k_t)$$

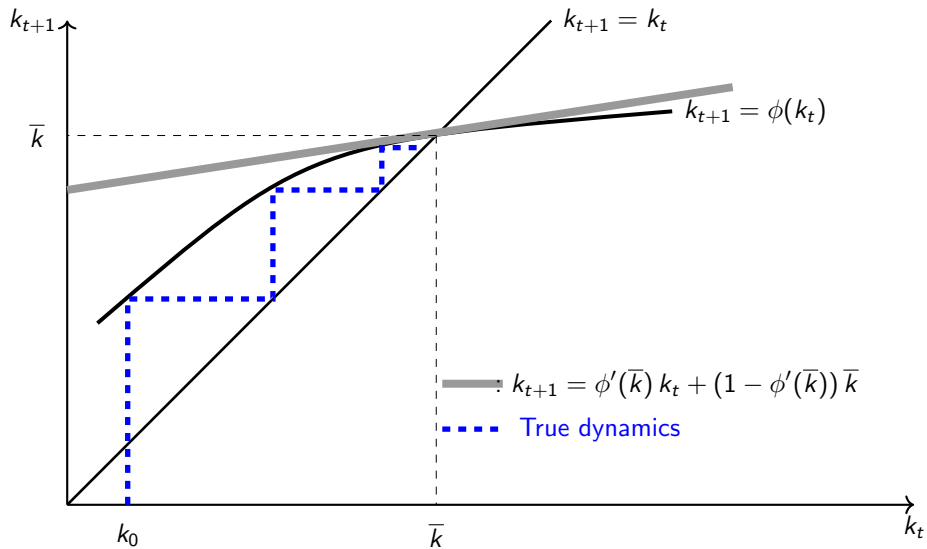
- ▶ The steady state is $\bar{k} = \varphi(\bar{k})$
- ▶ Linear approximation:

$$(k_{t+1} - \bar{k}) \approx \varphi'(\bar{k})(k_t - \bar{k})$$

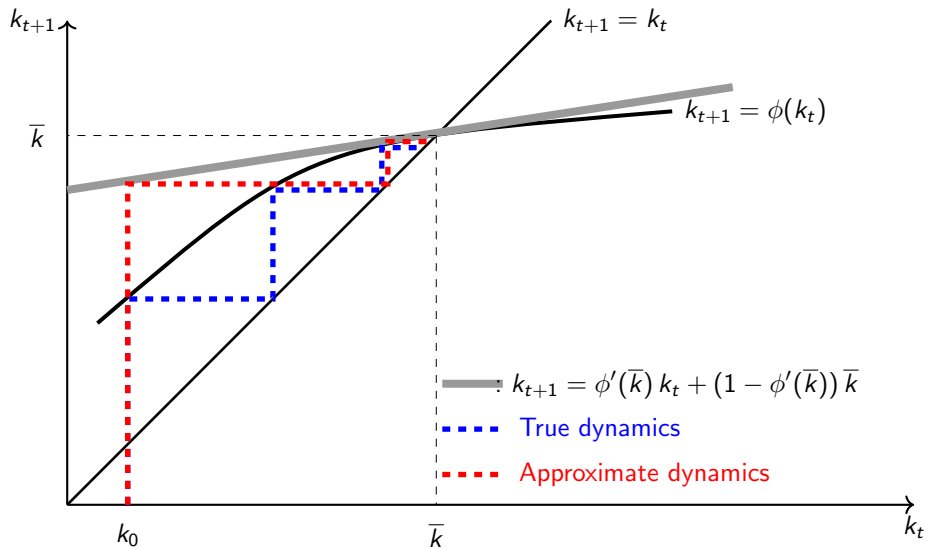
10. Linear approximation



10. Linear approximation



10. Linear approximation



10. Linear approximation

Solution

- ▶ Let's show an important result: the model solution can be partitioned into a feedback and an feedforward part.

- ▶ Model is

$$H(k_t, k_{t+1}, k_{t+2}, z_t, z_{t+1}) = 0$$

- ▶ The steady state is given by

$$H(\bar{k}, \bar{k}, \bar{k}, \bar{z}, \bar{z}) = 0$$

- ▶ Linear approximation is

$$\begin{aligned} & H_{k_t} \times (k_t - \bar{k}) + H_{k_{t+1}} \times (k_{t+1} - \bar{k}) + H_{k_{t+2}} \times (k_{t+2} - \bar{k}) \\ & + H_{z_t} \times (z_t - \bar{z}) + H_{z_{t+1}} \times (z_{t+1} - \bar{z}) \\ & = 0 \end{aligned} \tag{10.1}$$

with $H_{k_t} = H_{k_t}(\bar{k}, \bar{k}, \bar{k}, \bar{z}, \bar{z}), \dots$

10. Linear approximation

Solution

- Rewrite (10.1) as

$$\phi_0 k_{t+2} + \phi_1 k_{t+1} + \phi_2 k_t = A_0 + A_1 z_t + A_2 z_{t+1} \quad (10.2)$$

or

$$\phi(L)k_{t+2} = A_0 + A_1 z_t + A_2 z_{t+1} \quad (10.3)$$

- We want to solve this equation, i.e. find k_{t+1} as a function of past endogenous variables (k_{t-j}) and exogenous variables z (past, present or future as there is here no uncertainty)
- To do so, we will manipulate and transform the characteristic polynomial $\phi(L) = \phi_0 + \phi_1 L + \phi_2 L^2$
- Let $\mu_{1,2}$ be the two roots of ϕ (the solutions to $\phi(L) = 0$). Assume they are non-zero, real and distinct (can be proved in some environments)
- We have $\phi(L) = \phi_2(\mu_1 - L)(\mu_2 - L)$ and $\mu_1\mu_2 = \phi_0/\phi_2$.

10. Linear approximation

Solution

- Write $\mu_i - L = \mu_i \left(1 - \frac{1}{\mu_i} L\right)$ so that $\phi(L)$ can be written

$$\underbrace{\phi_2 \mu_1 \mu_2}_{\phi_0} \left(1 - \frac{1}{\mu_1} L\right) \left(1 - \frac{1}{\mu_2} L\right)$$

Denote $\lambda_i = \frac{1}{\mu_i}$ to obtain

$$\phi(L) = \phi_0 (1 - \lambda_1 L)(1 - \lambda_2 L)$$

- Let's assume (more on this later) that $|\lambda_1| > 1$, $|\lambda_2| < 1$

10. Linear approximation

Solution

- Because $|\lambda_1| > 1$,

$$(1 - \lambda_1 L)^{-1} = \sum_{j=0}^{\infty} \lambda_1^j L^j \quad \text{diverges}$$

- We can flip this infinite sum:

$$(1 - \lambda_1 L) = -\lambda_1 L (1 - \lambda_1^{-1} L^{-1})$$

and

$$(1 - \lambda_1^{-1} L^{-1})^{-1} = \sum_{j=0}^{\infty} \lambda_1^{-j} L^{-j} \quad \text{converges}$$

- Recall that $L^{-1}x_t = x_{t+1}$
- $\sum_{j=0}^{\infty} \lambda_1^{-j} L^{-j}$ is a forward looking term, which corresponds to a discounted sum of future values, with discounting at rate λ_1^{-1}

10. Linear approximation

Solution

- We can then rewrite $\phi(L)$ as follows:

$$\begin{aligned}\phi(L) &= \phi_0(1 - \lambda_1 L)(1 - \lambda_2 L) \\ &= \phi_0(-\lambda_1 L)(1 - \lambda_1^{-1} L^{-1})(1 - \lambda_2 L)\end{aligned}$$

- and using $\phi_2 = \lambda_1 \lambda_2 \phi_0$:

$$\phi(L) = \frac{-\phi_2}{\lambda_2} L (1 - \lambda_1^{-1} L^{-1})(1 - \lambda_2 L)$$

- so that (10.3)

$$\phi(L)k_{t+2} = A_0 + A_1 z_t + A_2 z_{t+1} \quad (10.3)$$

writes

$$\frac{-\phi_2}{\lambda_2} (1 - \lambda_1^{-1} L^{-1})(1 - \lambda_2 L) L k_{t+2} = A_0 + A_1 z_t + A_2 z_{t+1} \quad (10.6)$$

10. Linear approximation

Solution

$$\frac{-\phi_2}{\lambda_2} (1 - \lambda_1^{-1}L^{-1})(1 - \lambda_2L)Lk_{t+2} = A_0 + A_1z_t + A_2z_{t+1}$$

- Put the **blue term** on the right-hand side of the equation:

$$\underbrace{(1 - \lambda_2L)k_{t+1}}_{\substack{\text{Transient dynamics,} \\ \text{"feedback",} \\ \text{"backward looking"}}} = \underbrace{\frac{-\lambda_2\phi_2^{-1}}{(1 - \lambda_1^{-1}L^{-1})}A_0 + A_1z_t + A_2z_{t+1}}_{\substack{\text{Expectational dynamics,} \\ \text{"feedforward",} \\ \text{"forward looking"}}} \quad (10.7)$$

10. Linear approximation

Solution

- ▶ (10.7) can be more explicitly written as

$$k_{t+1} = \lambda_2 k_t - \lambda_2 \phi_2^{-1} \sum_{j=0}^{\infty} (\lambda_1)^{-j} [A_0 + A_1 z_{t+j} + A_2 z_{t+j+1}] \quad (10.8)$$

- ▶ $(\lambda_1)^{-j}$ is the rate at which expectations about the future are discounted
- ▶ The derivation relies on the fact that $|\lambda_1| > 1$ and $|\lambda_2| < 1$.
- ▶ Whether this is true or not depends on the economic environment.
- ▶ It is true in the neoclassical growth model we are working with.

10. Linear approximation

Relation with the shooting algorithm

- ▶ k_0 is given
- ▶ In the linearized model, k_1 (or equivalently c_0) is chosen looking at the whole future.
- ▶ It corresponds in the shooting algorithm to the choice of the c_0 such that $k_S = \bar{k}$ after S periods, i.e. *in the future*

10.1. Relation between the λ_i s

- When $\{g_t, \tau_t\} = 0 \ \forall t$, one can prove (a bit long) that

$$\lambda_1 \lambda_2 = 1/\beta$$

and that

$$|\lambda_1| > 1/\sqrt{\beta}$$

and

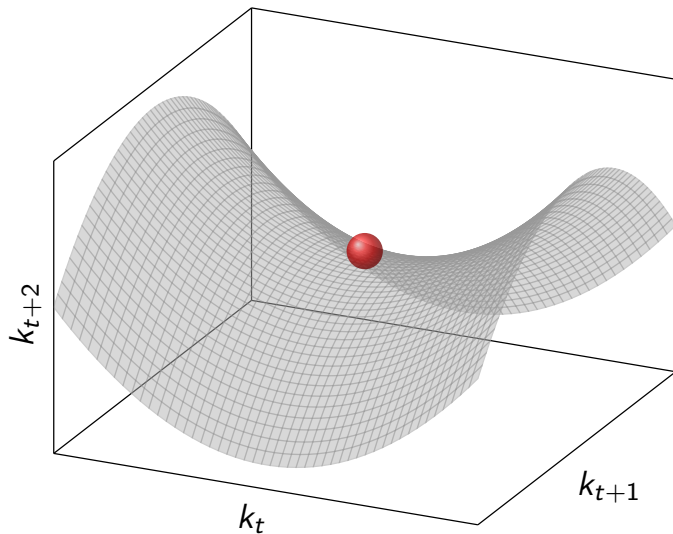
$$|\lambda_2| < 1/\sqrt{\beta}$$

10.2. Existence and uniqueness of the equilibrium dynamics

- ▶ When $|\lambda_1| > 1$ and $|\lambda_2| < 1$, we have existence and uniqueness of the equilibrium dynamics
- ▶ This is a case in which for given k_0 , there is a unique c_0 that satisfies non explosion.
- ▶ This is what we call *saddle-path stability*
- ▶ There are as many roots on the unit disc as predetermined variables = Blanchard-Kahn [1980] condition

10.2. Existence and uniqueness of the equilibrium dynamics

Saddle Path Stability



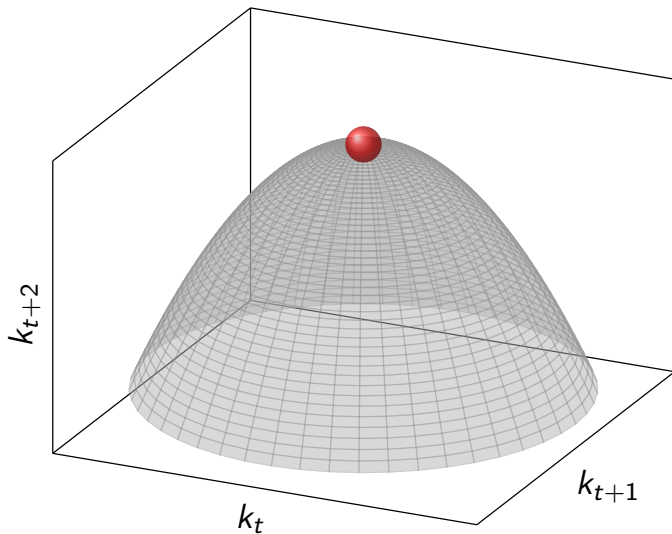
10.2. Existence and uniqueness of the equilibrium dynamics

Instability

- ▶ When $|\lambda_1| > 1$ and $|\lambda_2| > 1$, the model becomes explosive.
- ▶ One would need k_0 to jump to $\bar{k}$, but this is not possible as k_0 is predetermined.
- ▶ The economy will explode and at some point will violate resource constraint or positivity of c and k .
- ▶ The equilibrium does not exist.

10.2. Existence and uniqueness of the equilibrium dynamics

Instability



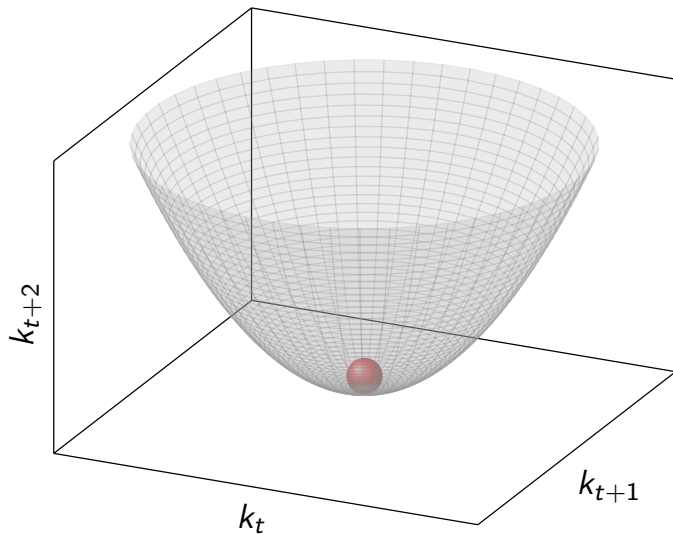
10.2. Existence and uniqueness of the equilibrium dynamics

Indeterminacy

- ▶ When $|\lambda_1| < 1$ and $|\lambda_2| < 1$, the model is *indeterminate*: there is a continuum of paths that converge to the steady state.
- ▶ Given k_0 , any c_0 is admissible.
- ▶ There are *sunspot equilibria*: if the economy believes that it should start from some $\tilde{c}_0$, this is an equilibrium, and many $\tilde{c}_0$ are admissible.

10.2. Existence and uniqueness of the equilibrium dynamics

Indeterminacy



10.3. Once-and-for-all jumps

- ▶ Given the above algebra, we can write the full approximate solution following a once-and-for-all jump in one forcing variable.
- ▶ Assume that the economy is initially at the steady state, that we normalize to $\bar{k} = \bar{z} = 0$
- ▶ Assume z is of dimension 1.
- ▶ The shock is :

$$z_t = \begin{cases} 0 & \text{if } t \leq T-1 \\ \tilde{z} & \text{if } t \geq T-1 \end{cases}$$

10.3. Once-and-for-all jumps

► Define:

$$v_t = \sum_{i=0}^{\infty} \lambda_1^{-i} z_{t+i} = \begin{cases} \left(\frac{1}{\lambda_1}\right)^{T-t} \frac{1}{1 - (\frac{1}{\lambda_1})} \tilde{z} & \text{if } t \leq T-1 \\ \frac{1}{1 - (\frac{1}{\lambda_1})} \tilde{z} & \text{if } t \geq T-1 \end{cases} \quad (10.10)$$

$$h_t = \sum_{i=0}^{\infty} \lambda_1^{-i} z_{t+i+1} = \begin{cases} \left(\frac{1}{\lambda_1}\right)^{T-(t+1)} \frac{1}{1 - (\frac{1}{\lambda_1})} \tilde{z} & \text{if } t \leq T-1 \\ \frac{1}{1 - (\frac{1}{\lambda_1})} \tilde{z} & \text{if } t \geq T-1 \end{cases} \quad (10.11)$$

10.3. Once-and-for-all jumps

► Then using

$$k_{t+1} = \lambda_2 k_t - \lambda_2 \phi_2^{-1} \sum_{j=0}^{\infty} (\lambda_1)^{-j} [A_0 + A_1 z_{t+j} + A_2 z_{t+j+1}] \quad (10.8)$$

we obtain the solution

$$k_{t+1} = \begin{cases} \lambda_2 k_t - \frac{(\phi_0 \lambda_1)^{-1} A_0}{1 - \frac{1}{\lambda_1}} - \frac{(\phi_0 \lambda_1)^{-1} (\frac{1}{\lambda_1})^{T-t}}{1 - \frac{1}{\lambda_1}} (A_1 + A_2 \lambda_2) \tilde{z} & \text{if } t \leq T-1 \\ \lambda_2 k_t - \frac{(\phi_0 \lambda_1)^{-1}}{1 - \frac{1}{\lambda_1}} (A_0 + A_1 + A_2 \lambda_2) \tilde{z} & \text{if } t \geq T-1 \end{cases} \quad (10.10)$$

11. Growth

- ▶ Now $Y_t = F(K_t, A_t n_t)$
- ▶ $A_{t+1} = \mu A_t$
- ▶ Deflate quantity variables: $y_t = \frac{Y_t}{A_t n_t}$, $k_t = \frac{K_t}{A_t n_t}$, $c_t = \frac{C_t}{A_t n_t}$, $g_t = \frac{G_t}{A_t n_t}$
- ▶ $y_t = f(k_t) = F(k_t, 1)$

11. Growth

- ▶ Assume again that labour is inelastically supplied and $n_1 = 1$
- ▶ Feasibility:

$$k_{t+1} = \mu^{-1}(f(k_t) + (1 - \delta)k_t - g_t - c_t) \quad (11.4)$$

- ▶ Euler:

$$u'(A_t c_t) = \beta u'(A_{t+1} c_{t+1}) \frac{(1 + \tau_{ct})}{(1 + \tau_{ct+1})} ((1 - \tau_{kt+1})(f'(k_{t+1}) - \delta) + 1) \quad (11.5)$$

- ▶ With $u = \frac{c^{1-\gamma}}{1-\gamma}$,

$$\left(\frac{c_{t+1}}{c_t}\right)^\gamma = \beta \mu^{-\gamma} \bar{R}_{t+1}$$

$\rightsquigarrow$ it is “as if” discount rate is now $\beta \mu^{-\gamma}$ $\rightsquigarrow$ growth increases discounting because marginal utility is decreasing (therefore future units of good are worth less with growth).

11. Growth

- At the steady state of the deflated economy (which corresponds to a *balanced growth path* of the non deflated economy):

$$f'(\bar{k}) = \delta + \left(\frac{(1 + \rho)\mu^\gamma - 1}{1 - \tau_k} \right)$$

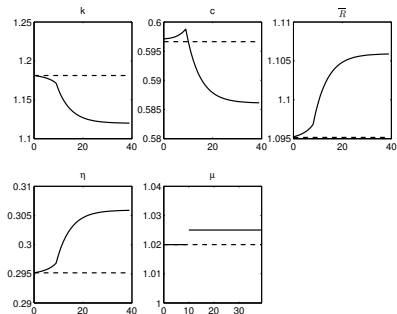
$\rightsquigarrow \bar{k}$ is smaller when $\mu > 1$ (as compared to $\mu = 1$)

11. Growth

- ▶ We can solve the deflated economy using the shooting algorithm
- ▶ Then we can recover the levels by multiplying by A_t : $K_t = A_t k_t = A_0 \mu^t k(t)$, etc...
- ▶ Note that a permanent increase in μ

11. Growth

Foreseen permanent increase in μ

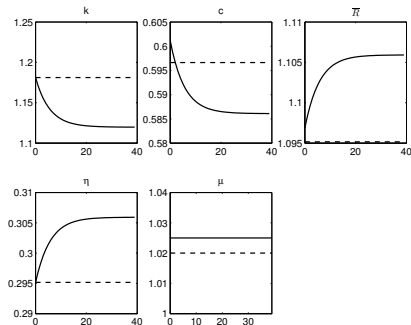


- ▶ New steady state level of k is lower
- ▶ Consumption jumps immediately because people are wealthier.
- ▶ Increase in the gross return $\bar{R}$

Figure 11.11.1: Response to foreseen once-and-for-all increase in rate of growth of productivity μ at $t = 10$. From left to right, top to bottom: $k, c, \bar{R}, \eta, \mu$, where now k, c are measured in units of effective unit of labor.

11. Growth

Surprise permanent increase in μ



- It looks very much like the transient part (after period 10) of the previous figure
- Increase in the gross return $\bar{R}$

Figure 11.11.2: Response to increase in rate of growth of productivity μ at $t = 0$. From left to right, top to bottom: $k, c, \bar{R}, \eta, \mu$, where now k, c are measured in units of effective unit of labor.

12. Elastic Labour supply

$$\max \mathcal{L} = \sum_{t=0}^{\infty} \beta^t U(c_t, 1 - n_t) + \mu \text{ ibc}$$

$$\sum_{t=0}^{\infty} q_t ((1 + \tau_{ct})c_t + (k_{t+1} - (1 - \delta)k_t)) \leq \sum_{t=0}^{\infty} q_t (\eta_t k_t - \tau_{kt}(\eta_t - \delta)k_t + (1 - \tau_{nt})w_t n_t - \tau_{ht}) \quad (\text{ibc})$$

- On top of the Euler equation, have an extra *foc*, which is the static consumption leisure decision.
- The two *foc* write

$$\begin{aligned} U_1(F(k_t, n_t) + (1 - \delta)k_t - g_t - k_{t+1}, 1 - n_t) &= \beta \left(\frac{1 + \tau_{ct}}{1 + \tau_{ct+1}} \right) \\ &\quad \times U_1(F(k_{t+1}, n_{t+1}) + (1 - \delta)k_{t+1} - g_t - k_{t+2}, 1 - n_{t+1}) \\ &\quad \times [(1 - \tau_{kt+1})(F_k(k_{t+1}, n_{t+1}) - \delta) + 1] \\ \frac{U_2(F(k_t, n_t) + (1 - \delta)k_t - g_t - k_{t+1}, 1 - n_t)}{U_1(F(k_t, n_t) + (1 - \delta)k_t - g_t - k_{t+1}, 1 - n_t)} &= \left(\frac{1 - \tau_{nt}}{1 + \tau_{ct}} \right) F_n(k_t, n_t) \end{aligned}$$

12. Elastic Labour supply

Steady state

- ▶ We can again solve the model using the shooting algorithm or solving a linearized version.
- ▶ The steady state is now given by

$$\beta(1 + (1 - \tau_k)(F_k(\bar{k}, \bar{n}) - \delta)) = 1 \quad (12.5)$$

$$\frac{U_2(\bar{c}, 1 - \bar{n})}{U_1(\bar{c}, 1 - \bar{n})} = \left(\frac{1 - \tau_n}{1 + \tau_c} \right) F_n(\bar{k}, \bar{n}) \quad (12.6)$$

$$\bar{c} + \bar{g} + \delta \bar{k} = F(\bar{k}, \bar{n}) \quad (12.7)$$

- ▶ Given that $F_k(\bar{k}, \bar{n}) = F_k(\frac{\bar{k}}{\bar{n}}, 1)$, (12.5) pins down $\tilde{k} = \frac{\bar{k}}{\bar{n}}$
- ▶ (12.7) writes

$$\delta + \frac{\rho}{1 - \tau_k} = f(\tilde{k})$$

$\rightsquigarrow$ only τ_k distorts $\tilde{k}$.

- ▶ But τ_c and τ_n now distort the consumption/leisure decision.

12. Elastic Labour supply

Steady state

- ▶ Assume $U(c, 1 - n) = \log c + B \times (1 - n)$ (Hansen-Rogerson preferences)
- ▶ B is chosen such that $0 < \bar{n} < 1$
- ▶ $\tilde{k}$ can be computed from $f(\tilde{k}) = \delta + \frac{\rho}{1 - \tau_k}$
- ▶ The rest of the steady state can be computed as follows:
 - × (12.6) implies $\bar{c} = \frac{1}{B} \left(\frac{1 - \tau_n}{1 + \tau_c} \right) (f(\tilde{k}) - \tilde{k}(f'(\tilde{k})))$
 - × Then (12.7) implies $\bar{c} + \bar{g} + \delta\bar{k} = \bar{n}f(\tilde{k})$ so that

$$\bar{n} = (f(\tilde{k}) - \delta\tilde{k})^{-1}(\bar{c} + \bar{g}) \quad (12.14)$$

which pins down $\bar{n}$

- × Once $\bar{n}$ and $\tilde{k}$ are known, $\bar{k} = \bar{n}\tilde{k}$ can be obtained
- ▶ Let's assume same parameters values plus $B = 3$.

12. Elastic Labour supply

Unforeseen permanent increase in g

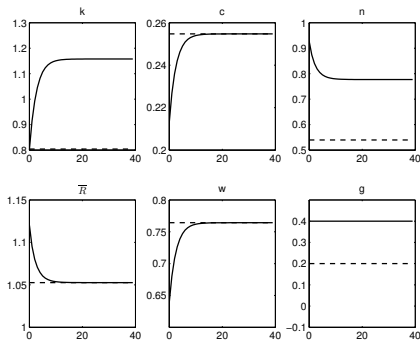
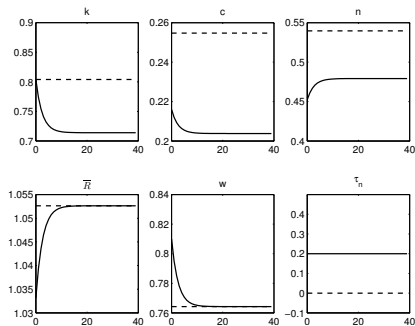


Figure 11.12.1: Elastic labor supply: response to unforeseen increase in g at $t = 0$. From left to right, top to bottom: $k, c, n, \bar{R}, w, g$. The dashed line is the original steady state.

- We have shown that $\bar{k}/\bar{n}$ and $\bar{c}$ are not affected at the steady state
- (12.14) then implies that $\bar{n} \nearrow$ and therefore that $\bar{k} \nearrow$
- In the transition, $c \searrow$ and $n \nearrow$, which is bad for welfare.

12. Elastic Labour supply

Unforeseen permanent increase in τ_n



► Labour is discouraged $\rightsquigarrow$ the economy shrinks

Figure 11.12.2: Elastic labor supply: response to unforeseen increase in τ_n at $t = 0$. From left to right, top to bottom: $k, c, n, \bar{R}, w, \tau_n$. The dashed line is the original steady state.

12. Elastic Labour supply

Foreseen permanent increase in τ_n

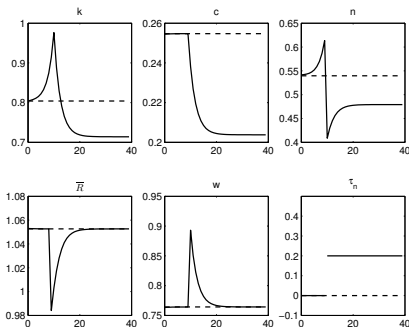


Figure 11.12.3: Elastic labor supply: response to foreseen increase in τ_n at $t = 10$. From left to right, top to bottom: $k, c, n, \bar{R}, w, \tau_n$. The dashed line is the original steady state.

- ▶ Long run effects are the same
- ▶ But in the short run $n, k \nearrow$ while c is flat
- ▶ It is worth working more (and saving) while labour is less taxed (before period 10)
- ▶ The impact of unexpected vs expected tax increase is in line with what is found in the data.
- ▶ MERTENS and RAVN [2011], “Understanding the Effects of Anticipated and Unanticipated Tax Policy Shocks.” Review of Economic Dynamics 14(1): 27-54. (Effect of tax cuts)

12. Elastic Labour supply

The Response to Tax Cuts in the US – Anticipated tax cuts are announced at date -6 and implemented at date 0 (MERTENS and RAVN [2011])

